Coordinated World Model Learning for Robot Teams: 2.5D Terrain Enhancement with NASA DEM Data

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ML Course Project – 2026

Abstract

This project implements a multi-robot World Model (WM) learning framework enhanced with NASA 2.5D Digital Elevation Models (DEMs) in .IMG format. Building on the theoretical foundation of Skulimowski (2026), we extend flat 2D terrain representations to 2.5D geometry, enabling more accurate traversability estimation, physics-based velocity prediction, and 3D-aware path planning. The system simulates a team of three rovers and a lander, sharing WMs via communication graphs and using multi-criteria A* planning on real elevation data. We demonstrate that 2.5D terrain significantly improves each stage of the Decision Making Cycle (DMC), from feature detection to inference rules.

1 Introduction

Deep-space missions to icy moons such as Europa and Enceladus face communication delays and unknown environments, requiring autonomous robot teams. A central component is the **World Model (WM)** — a unified framework integrating SLAM, machine learning, sensors, and knowledge bases. The base paper by Skulimowski [1] defines a formal WM as:

$$WM(R, E, t) = (MM(R),\ RTM(R, E, t),\ X,\ g)$$

where $MM(R)$ is a meta-model, RTM is the real-time model, X is a decision engine, and g is a generalization function.

While the original work focuses on flat 2D imagery, this project introduces **2.5D terrain** (gridded elevation data) from NASA HiRISE DEMs. The goal is to show how 3D geometry improves team coordination, traversability assessment, and path planning.

2 Project Structure and Implementation

The Python implementation follows the architecture described in Table 1.

A synthetic DEM is used by default; real NASA .IMG files can be placed in `data/`.

3 Mathematical Enhancements from 2.5D Terrain

3.1 Slope and Roughness

From a DEM $Z(x, y)$, slope magnitude is:

$$\nabla Z(x, y) = \sqrt{\left(\frac{\partial Z}{\partial x}\right)^2 + \left(\frac{\partial Z}{\partial y}\right)^2}$$

Table 1: Project modules and their functions

Module	**Purpose**
`dem_loader.py`	Parse NASA .IMG (PDS) DEMs
`terrain_analysis.py`	Compute slope, roughness, traversability, velocity
`world_model.py`	Classes for MM, RTM, X, g
`robot.py`	Robot agent with WM, sensors, navigation
`path_planner.py`	Multi-criteria A* on 2.5D grid
`communication.py`	NetworkX communication graphs
`team_simulation.py`	Algorithms 1 & 2 from the paper

Slope angle $\theta = \arctan(\nabla Z)$ in degrees. Local roughness is the standard deviation of elevation in a 5×5 window:

$$R(x, y) = \sqrt{\mathrm{Var}(Z_{5\times 5})}$$

3.2 Traversability Label L_1

$$L_1(x, y) = 1 - \left[\alpha \frac{\theta}{\theta_{\max}} + \beta \frac{R}{R_{\max}}\right]$$

with $\alpha = 0.6$, $\beta = 0.4$, $\theta_{\max} = 45°$, $R_{\max} = 0.5\,\mathrm{m}$. $L_1 \in [0, 1]$ — higher is more traversable.

3.3 Physics-Based Velocity

$$V = V_{\max} \cdot f_{\text{slope}} \cdot f_{\text{roughness}}, \quad f_{\text{slope}} = \max(0, 1 - \frac{\theta}{45°}), \quad f_{\text{roughness}} = \max(0, 1 - \frac{R}{0.5})$$

where $V_{\max} = 2.0\,\mathrm{m/s}$.

3.4 3D-Aware Damage Cost

For an edge between two cells:

$$\text{cost} = (1 - \overline{L_1}) \cdot d_{3D} \cdot \beta_{\text{safety}}$$

where $d_{3D} = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$ (Euclidean 3D distance) and $\beta_{\text{safety}} = 5$.

4 Impact on the Decision Making Cycle (DMC)

The 2.5D enhancement improves every stage of the DMC, as summarized in Table 2.

Table 2: 2.5D contribution to each DMC stage

DMC stage	**2.5D benefit**
f_1: Feature detection	Edge detection from slope gradients $\partial(\nabla Z)/\partial x$
f_2: Object identification	Ridges from high-curvature terrain
f_3: Entity recognition	Craters, peaks from unambiguous topology
f_4: Label assignment	L_1 directly from slope, not heuristics
Ψ: Inference engine	High-confidence inputs for if-then rules

5 Simulation and Key Results

Running `run_simulation.py` or `run_demo_synthetic.py` produces:

- **Traversability map** — per-cell L_1 over the full DEM.
- **Path comparison** — flat 2D cost vs. 2.5D-aware cost (Fig. 1).
- **Communication graph evolution** — how robots share WM data.
- **Scientific yield** — quality indicators H_1, H_2 over time.
- **Velocity heatmap** — physics-based speed across terrain.

`robot_paths_2d_vs_25d.png`

Figure 1: Illustrative output: 2D vs 2.5D path planning. The 2.5D-aware planner avoids steep slopes and rough areas, even if the Euclidean distance is longer.

6 Conclusion and Future Work

This project demonstrates that integrating NASA 2.5D DEM data into a multi-robot WM framework yields:

1. More realistic traversability estimates.
2. Physics-based velocity that accounts for slope and roughness.
3. 3D-aware damage costs, improving path safety.
4. Higher precision in the Decision Making Cycle.

Future work includes slimming studies (minimum viable resolution), real-time sensor fusion (LiDAR + vision), and physical experiments with terrestrial rover teams.

Acknowledgments

The author thanks prof. dr hab. inż. Andrzej M. Skulimowski for the original WM framework and supervision. NASA HiRISE DEM data are publicly available at `https://www.uahirise.org/dtm/index.php`.

References

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